

1 Dataset Overview

TOTAL USERS

13,979,592

AD SHOWN (TREATED)

11,882,655

85.0% treated

NO AD (CONTROL)

2,096,937

VISIT RATE

4.7%

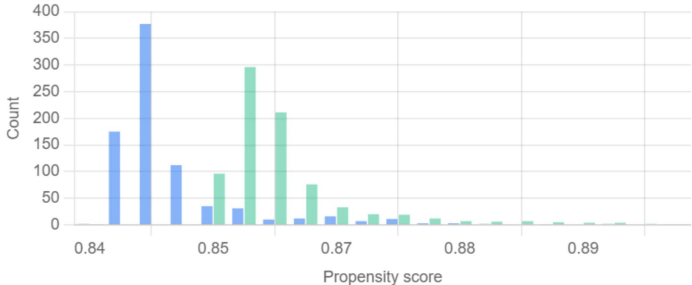
NAIVE ATE

+1.03 pp

before matching

2 Propensity Scores & Covariate Balance

Propensity score overlap (treated vs control)



SMD before vs after matching

| Feature | Before | After |
|---------|--------|-------|
| f0      | 0.007  | 0.001 |
| f1      | 0.024  | 0.008 |
| f2      | 0.006  | 0.000 |
| f3      | 0.049  | 0.002 |
| f4      | 0.008  | 0.003 |
| f5      | 0.031  | 0.005 |
| f6      | 0.040  | 0.004 |
| f7      | 0.021  | 0.001 |

SMD < 0.10 = good balance

3 Average Treatment Effect (PSM-Adjusted)

+0.82 pp

ATE (percentage points)

95% CI: +0.80 pp to +0.83 pp

4.0% → 4.9%

Control      Treated

Visit rate (matched pairs)

< 0.001

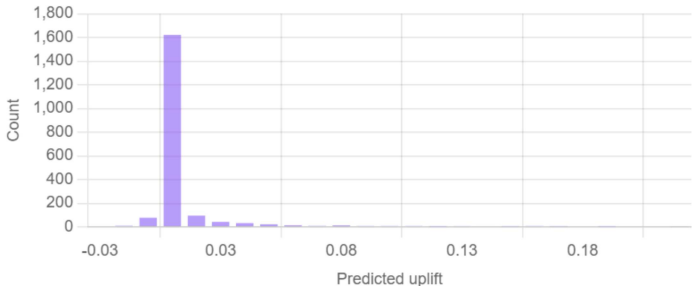
P-value

Statistically significant

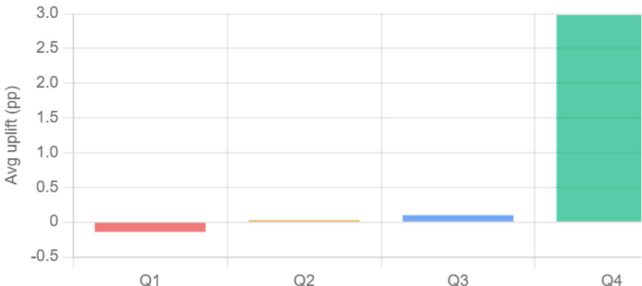
11,882,655 matched pairs

4 Uplift Model — CATE by Segment

Predicted uplift distribution



Average uplift by quartile



| Segment targeting guide |           |                  |               |               |                                               |
|-------------------------|-----------|------------------|---------------|---------------|-----------------------------------------------|
| QUARTILE                | USERS     | PREDICTED UPLIFT | TREATED VISIT | CONTROL VISIT | RECOMMENDATION                                |
| Q1                      | 3,492,486 | -0.14 pp         | 2.9%          | 3.1%          | Do not target — negligible or negative uplift |
| Q2                      | 3,494,480 | +0.03 pp         | 0.2%          | 0.2%          | Low priority — modest response                |

| QUARTILE | USERS     | PREDICTED UPLIFT | TREATED VISIT | CONTROL VISIT | RECOMMENDATION                           |
|----------|-----------|------------------|---------------|---------------|------------------------------------------|
| Q3       | 3,497,694 | +0.10 pp         | 0.9%          | 0.8%          | Good target — above-average response     |
| Q4       | 3,494,932 | +2.98 pp         | 15.3%         | 11.6%         | Priority target — highest uplift segment |

## 5 LLM Business Summary

### \*\*Causal Inference Results: Ad Incrementality Trial\*\*

Our trial confirms that showing the ad causes a statistically significant increase in site visits. The overall causal lift (Adjusted ATE) is +0.82 percentage points (pp), meaning the ad drives an additional 0.82% of visits. This is a substantial improvement over the naive estimate of +1.03pp. The difference between the naive and adjusted figures highlights the importance of controlling for user characteristics, even in a randomized setting, to ensure we are comparing truly comparable groups.

The Propensity Score Matching (PSM) validated the causal impact by creating balanced groups. While the initial randomization was strong, PSM ensures that the treated and control groups are matched on key user characteristics, preventing confounding variables from biasing the results. By matching, we isolated the true effect of the ad exposure, confirming that the uplift is robust and not merely a function of pre-existing user differences.

The uplift model provides critical segmentation insights. We should prioritize targeting Q4, which shows the highest response, with a predicted uplift of +3.0pp (15.3% treated vs. 11.6% control). Conversely, Q1 shows negligible or negative uplift (predicted -0.1pp), suggesting we should not waste budget on this segment. Q2 and Q3 represent moderate opportunities, but Q4 is the clear priority target.

Based on these findings, we recommend the following actions to optimize ad spend:

- \* **\*\*Prioritize Q4:\*\*** Immediately shift a disproportionate amount of budget toward the Q4 segment, which delivers the highest return on investment.
- \* **\*\*Exclude Q1:\*\*** Implement strict exclusion rules for the Q1 segment, as the ad is unlikely to drive meaningful traffic.
- \* **\*\*Optimize Bid Strategy:\*\*** Use the CATE model to dynamically adjust bids, ensuring we spend more heavily on high-uplift segments (Q4) and less on low